

DEEP LEARNING LAB

EXPERIMENT 2

Covariance, Eigenvalues and Eigenvectors Analysis

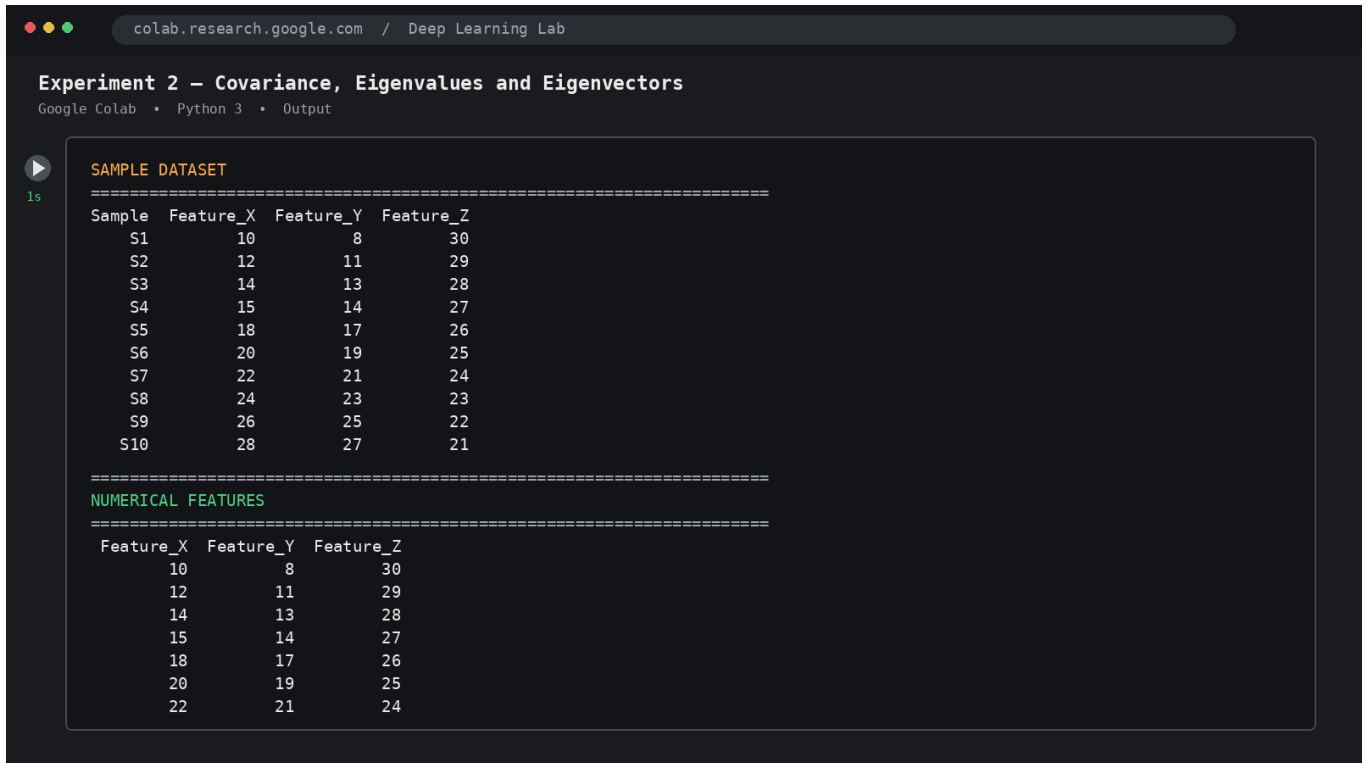


Figure 1: Sample Dataset and Numerical Features

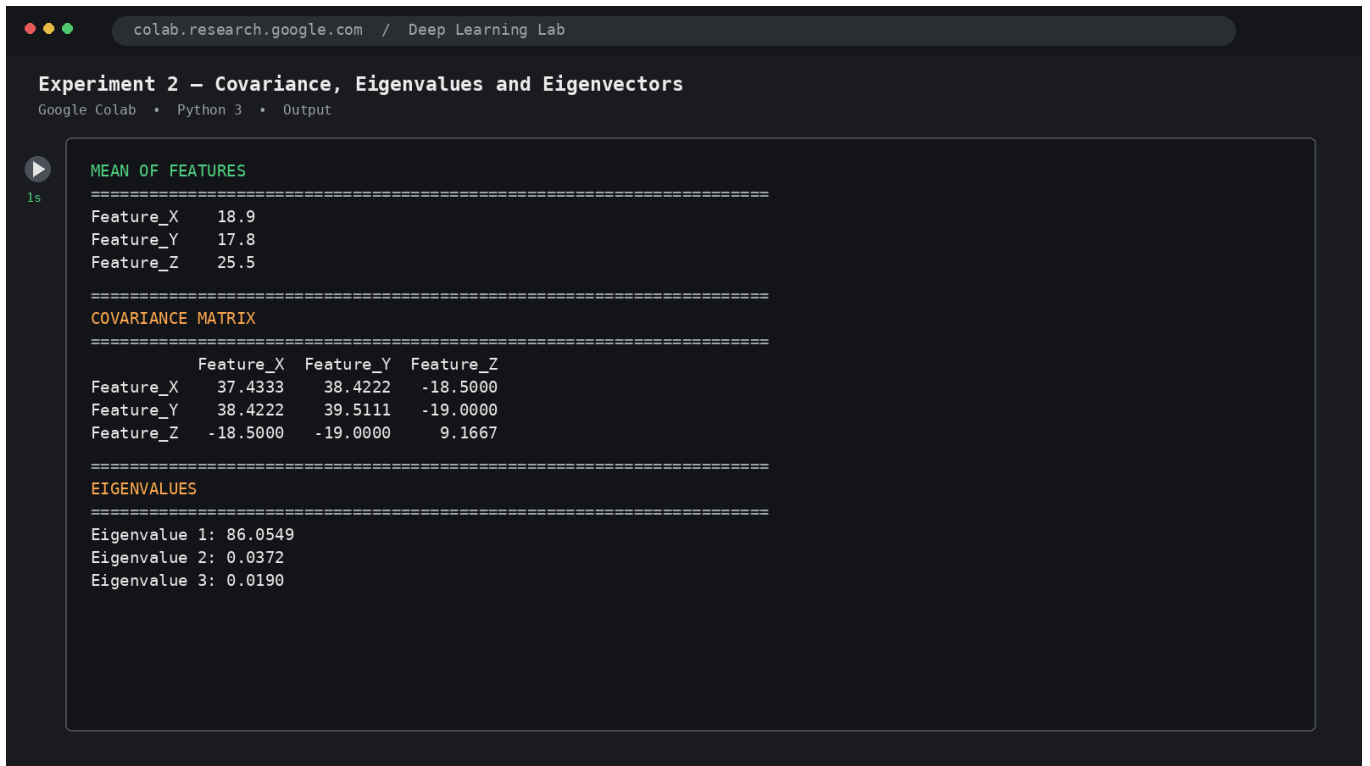


Figure 2: Mean, Covariance Matrix and Eigenvalues

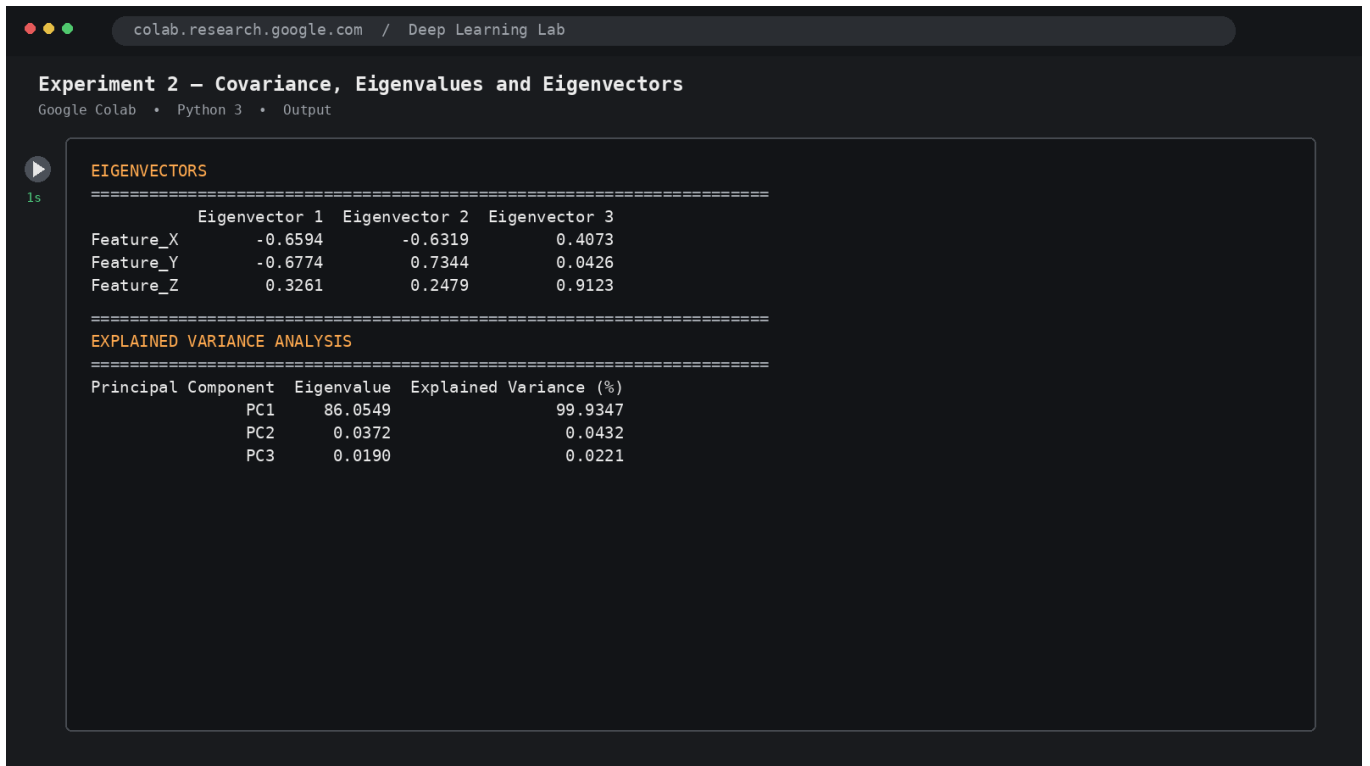


Figure 3: Eigenvectors and Explained Variance Analysis

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Experiment 2 – Covariance, Eigenvalues and Eigenvectors
Google Colab • Python 3 • Output

1s STATISTICAL EVALUATION
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Total Variance: 86.1111
Maximum Eigenvalue: 86.0549
Minimum Eigenvalue: 0.019
Highest Variance Component: PC1
PC1 Explained Variance: 99.93 %

FINAL RESULT
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Covariance, eigenvalues and eigenvectors were successfully
calculated using NumPy and Pandas.

The experiment demonstrated:
1. Sample dataset preparation
2. Mean calculation
3. Covariance matrix calculation
4. Eigenvalue calculation
5. Eigenvector calculation
6. Explained variance analysis
7. Statistical evaluation
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Figure 4: Statistical Evaluation and Final Result

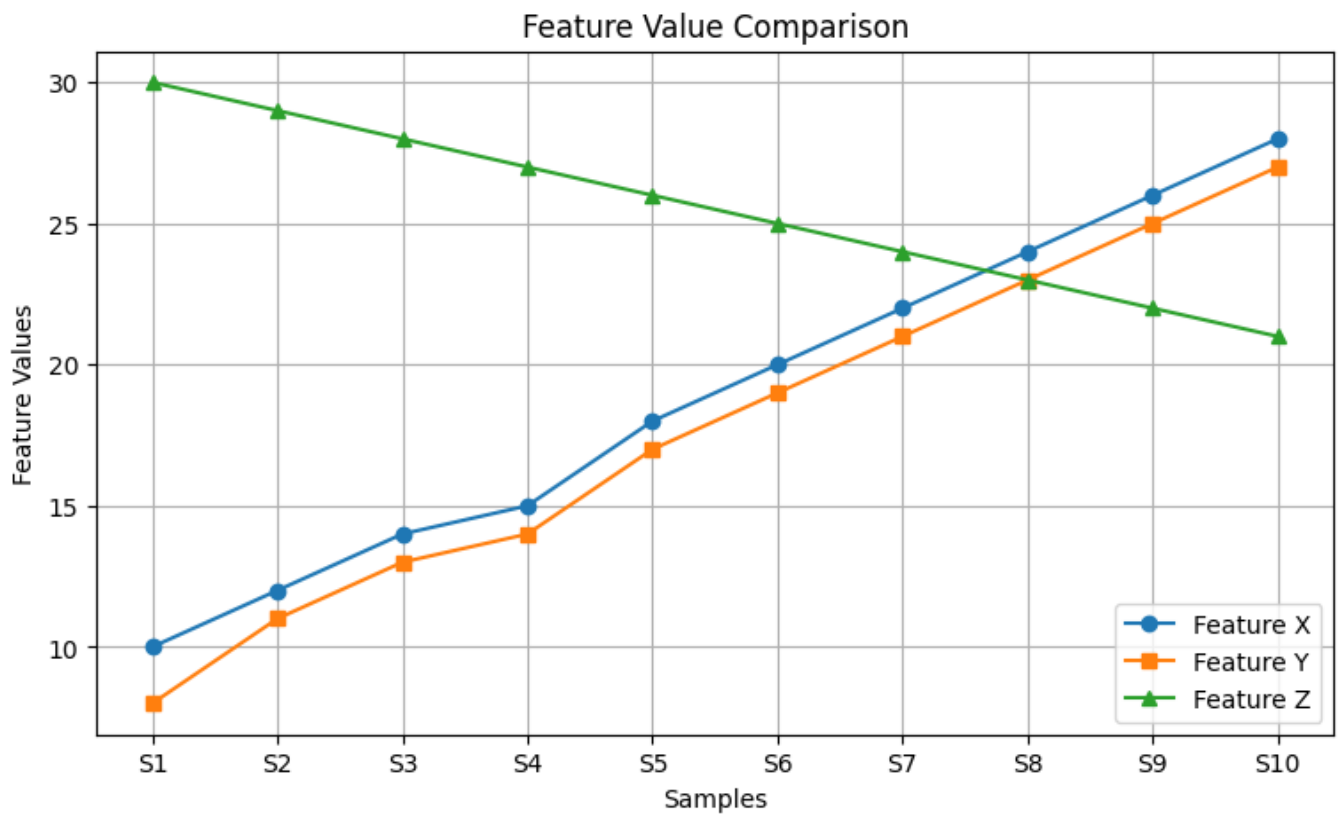


Figure 5: Feature Value Comparison

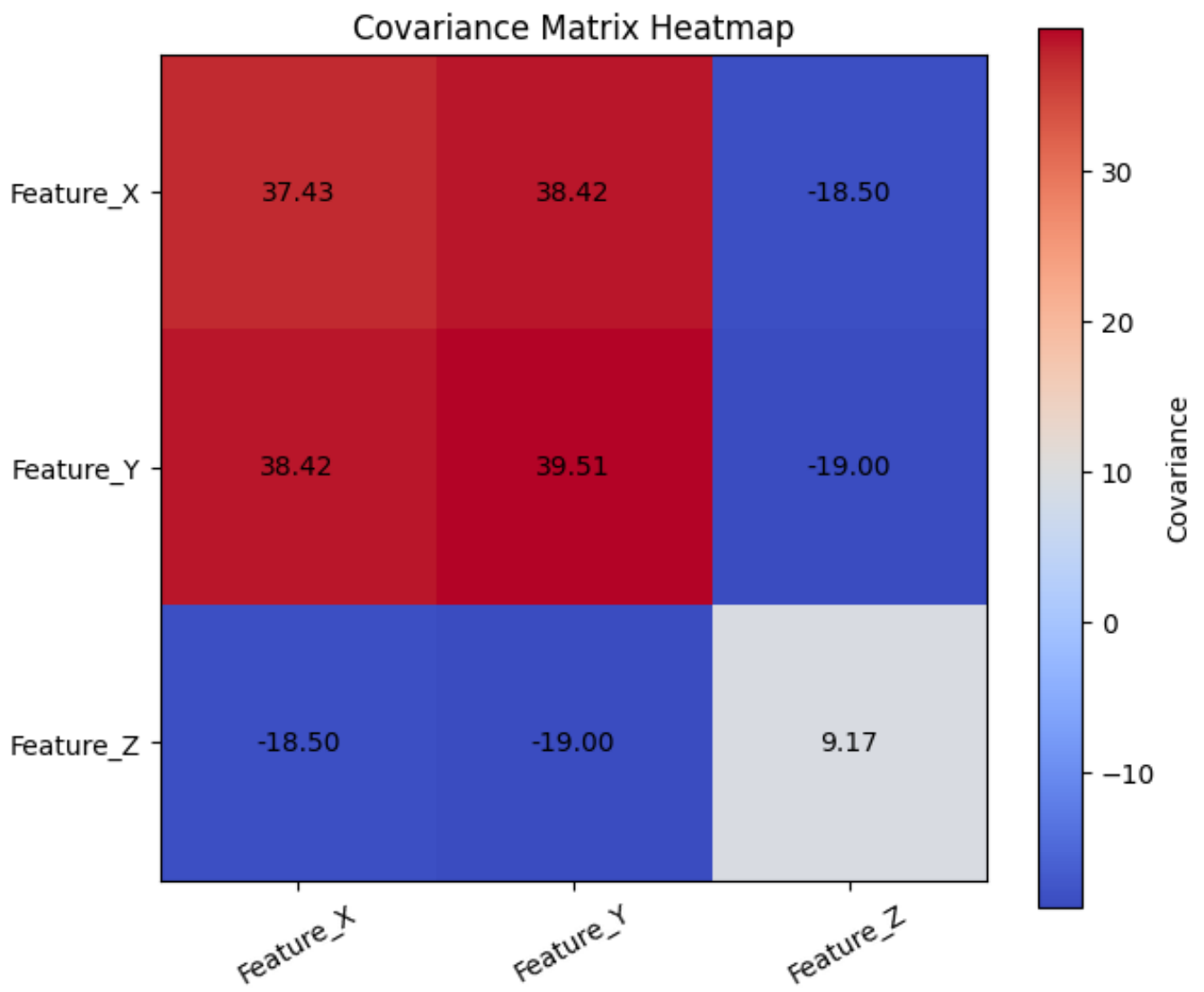


Figure 6: Covariance Matrix Heatmap

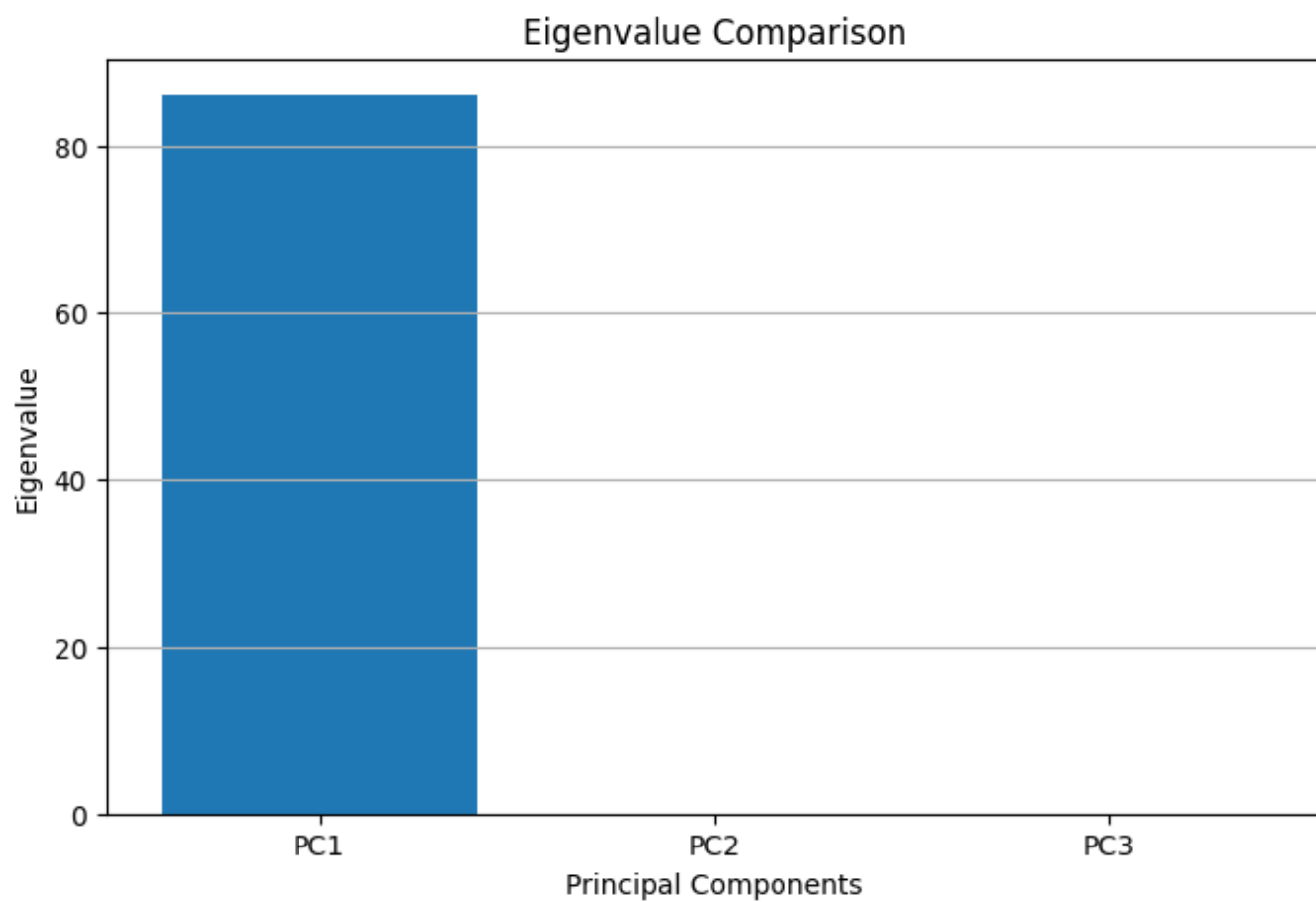


Figure 7: Eigenvalue Comparison